

WINNET CONSTRUCTION LTD • GHANA

PROJECT ARCHITECTURE & MASTER CODEBOOK

Complete System Blueprint, File Breakdown, Component Reference,
and Developer Customization Handbook

CORE SPECIFICATIONS

Frontend Framework: React 19 + TypeScript (Strict Type Safety)
Full-Stack Engine: TanStack React Start (SSR + Client Routing)
Styling & Design System: Tailwind CSS v4 + Motion Animations
Component Architecture: Radix UI Headless Primitives + Lucide Icons
SEO & Structured Data: Google JSON-LD GeneralContractor + PWA Manifest
Target Market: Ghana (Accra, Tema, Kumasi, Takoradi, Diaspora)
Version: 2.0.0 (Production Master)

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🔧 How to Use This Handbook

This document serves as your ultimate guide to maintaining, editing, extending, and deploying the Winnet Construction website. Every file is explained with its exact location, function, and modification recipes. You do not need deep programming knowledge to make changes like phone numbers, project photos, or text — just follow Section 8!

1. EXECUTIVE SUMMARY & TECH STACK

Modern high-performance web architecture

The Winnet Construction Ltd web application is engineered as an enterprise-grade digital platform designed to establish structural authority, showcase craftsmanship, and convert visitors into qualified project leads across Ghana and the diaspora.

Core Technologies Explained

- **React 19 & TypeScript:** Provides component modularity, instant UI updates, and strict compile-time type safety to prevent runtime crashes.
- **TanStack React Start & TanStack Router:** A modern, high-performance full-stack meta-framework providing Server-Side Rendering (SSR) for fast initial load times and client-side routing for instantaneous page transitions without full page reloads.
- **Tailwind CSS v4:** Next-generation styling engine providing fluid responsive grids, dark blueprint aesthetics, and custom brand utility tokens.
- **Motion (Framer Motion Engine):** Smooth, hardware-accelerated scroll reveal animations (`<Reveal />`) that give the website a premium, tactile architectural feel.
- **Lucide React Icons:** Over 40 crisp vector icons representing civil engineering, masonry, structural reinforcement, safety helmets, and communication channels.
- **Sharp & Node Image Generation Pipeline:** Automated asset processing script (`scripts/generate-icons.js`) that produces multi-resolution favicon and PWA icons (16px, 32px, 48px, 180px, 192px, 512px) from vector source SVG.
- **Google JSON-LD Structured Data:** Comprehensive Schema.org integration (`GeneralContractor`, `WebSite`, `BreadcrumbList`) ensuring Google recognizes the business name, logo, phone number, and service areas.

2. DIRECTORY & FILE STRUCTURE BLUEPRINT

Organization of the codebase

Here is the architectural overview of the root and source directory tree:

3. CENTRAL CONFIGURATION HUB

File: `src/config/site.ts` — *The Most Important File*

`src/config/site.ts` is the single source of truth for all business information, contacts, projects, services, statistics, and FAQs. Changing data in this file automatically updates the entire website, including navigation, footers, JSON-LD schemas, and contact forms.

1. Company Information Object (`company``)

Contains core business identity, phone numbers, email, and motto:

```
export const company = {
  name: "Winnet Construction Ltd",
  shortName: "Winnet",
  tagline: "Building with Integrity, Precision, and Purpose.",
  motto: "Quality Built, Value Delivered",
  phone: "0549074200", // Local Ghanaian dial format
  phoneInternational: "233549074200", // International dial format (no leading +)
  phoneFormatted: "+233 54 907 4200", // Display formatted phone string
  whatsapp: "233549074200", // WhatsApp target number
  whatsappPrefill: "Hello Winnet Construction...", // Default chat message
  email: "Fredmawuli123@gmail.com", // Primary enquiry & lead inbox
  officeAddress: "Accra, Ghana",
  serviceAreas: ["Accra", "Tema", "Kumasi", "Takoradi", "Eastern Region", "Diaspora"],
  workingHours: "Monday - Saturday: 8:00 AM - 6:00 PM",
};
```

2. Portfolio Projects Collection (`projects``)

An array of project objects powering `/projects`` and dynamic `/projects/$slug`` case studies. Each project has:

- **id & slug:** Unique URL identifier (e.g. `villa-east-legon`` -> `/projects/villa-east-legon``)
- **title & category:** Project heading and category (`Residential``, `Commercial``, `Structural``, `Renovation``)
- **location & duration:** Geographic location in Ghana and construction timeline
- **summary & scope:** Executive summary and array of structural deliverables
- **beforeAfter:** Optional before/after comparison imagery

3. Construction Services (`services``)

Defines all 6 specialized service categories (Residential Building, Commercial Complexes, Structural Framing, Modern Renovations, Architectural Finishes, Construction Consultation & BOQ) with icon keys, deliverables, and summaries.

4. Value Metrics (`stats``) & FAQs (`faqs``)

Contains key social proof numbers (Projects Completed, Client Satisfaction, Active Sites) and structured accordion Q&As addressing common questions from local and diaspora clients.

4. ROUTING ARCHITECTURE & PAGE ANATOMY

TanStack Router File-System Routing

The routing system in `src/routes/` is file-based and type-safe. Every page exports a TanStack `Route` created with `createFileRoute('/path')`.

Summary of Application Routes:

- **`__root.tsx`**: Global root layout. Injects `<HeadContent />`, SEO schemas, Google site verification, `<Navbar />`, `<EnquiryProvider />`, `<WhatsAppButton />`, `<Footer />`, and global `<Toaster />`.
 - **`index.tsx` (`/`)**: Homepage. Features high-impact Hero, Trust Bar, Core Services grid, Featured Projects showcase, Why Choose Winnet grid, Process overview, and Lead Enquiry CTA.
 - **`about.tsx` (`/about`)**: Company story, core engineering values (Honesty, Structural Durability, Quality Control, Timeliness), team philosophy, and on-site standards.
 - **`services.tsx` (`/services`)**: Full catalog of building services with scope checklists, process steps, and direct quote request triggers.
 - **`projects.tsx` (`/projects`)**: Portfolio page featuring category filter tabs (All, Residential, Commercial, Structural, Renovation) and interactive project cards.
 - **`projects.$slug.tsx` (`/projects/:slug`)**: Deep-dive case study page dynamically loaded by URL slug. Displays high-resolution gallery, project timeline, location, structural scope checklist, and inquiry box.
 - **`process.tsx` (`/process`)**: Explains the 6-step building roadmap: 1. Consultation -> 2. BOQ & Estimation -> 3. Architectural Planning -> 4. Foundation & Structure -> 5. Finishing -> 6. Handover.
 - **`why-winnet.tsx` (`/why-winnet`)**: Details the Winnet Advantage: Certified engineers, diaspora remote project tracking, transparent material sourcing, and zero hidden costs.
 - **`gallery.tsx` (`/gallery`)**: Visual on-site photography grid categorized by masonry, rebar reinforcement, concrete pours, and architectural interior finishes with interactive image lightbox.
 - **`contact.tsx` (`/contact`)**: Full contact desk featuring phone, email, WhatsApp, office hours, location map placeholder, and the interactive Project Estimation & Quote Form.
-

5. COMPONENT HIERARCHY & INTERACTIVE MODULES

Reusable building blocks

1. Layout Components (`src/components/layout/`)

- **`Navbar.tsx`**: Sticky header with animated blur backdrop, desktop navigation links, mobile hamburger drawer, direct phone click-to-call button, and 'Get a Quote' CTA button.
- **`Footer.tsx`**: Comprehensive footer containing company overview, quick links, service catalog links, direct WhatsApp click, business hours, and copyright disclaimer.
- **`WhatsAppButton.tsx`**: Floating bottom-right WhatsApp launcher with glowing pulse animation, opening an instant pre-filled chat with Winnet's official WhatsApp number.
- **`PageHeader.tsx`**: Standardized dark blueprint hero banner for inner pages with breadcrumb navigation and subtle grid overlay.

2. Project Enquiry System (`src/components/enquiry/`)

- **`EnquiryProvider.tsx`**: React Context provider allowing any button on any page to trigger the quote modal using `const { openEnquiry } = useEnquiry();``.
- **`ProjectEnquiryForm.tsx`**: Interactive multi-field estimation form with project type selection (Residential, Commercial, Renovation, Structural), location input, budget range, drawing attachment note, and automatic WhatsApp/Email routing.

3. Visual & Animation Components

- **`Reveal.tsx`**: Lightweight animation wrapper using Motion. Fades and slides elements into view smoothly as the user scrolls down the page.
- **`WinnetBrand.tsx`**: Crisp vector SVG logos including horizontal header lockup, square gold monogram badge, and footer marks.

6. DESIGN SYSTEM, COLORS & TYPOGRAPHY

File: `src/styles.css` — Visual Identity

The visual identity combines industrial durability with luxury architectural craft:

```
/* Core Color Tokens */
--color-gold: #F2B23C;           /* Signature Construction Gold Accent */
--color-gold-dark: #C98E20;      /* Deep Gold for hover states */
--color-gold-light: #FDE8B3;     /* Subtle Gold Tint for badge backgrounds */
--color-ink: #0D0D0D;            /* Deep Architectural Blueprint Charcoal/Black */
--color-ink-muted: #1A1A1A;     /* Elevated Dark Card Surface */
--color-ink-subtle: #2A2A2A;     /* Card Border & Divider Lines */
--color-bone: #F7F5F0;          /* Warm Off-White Light Section Background */
--color-steel: #71717A;         /* Secondary Text / Engineering Zinc Gray */
```

Custom CSS utility classes in `src/styles.css` include `.shell` (centered max-width container with responsive padding), `.h-display` (bold display typography), `.eyebrow` (tracked uppercase section labels), and `.blueprint-grid-dark` (subtle structural grid pattern).

7. SEO ENGINE, STRUCTURED DATA & FAVICONS

Maximum Google visibility & PWA support

1. Schema.org JSON-LD Structured Data (`src/lib/seo.ts`)

Injected into the `` of the root layout to inform search engines of your business credentials:

- **`WebSite` Schema:** Declares the official site name 'Winnet Construction Ltd' and alternate brand names for Google search snippet headers.
- **`GeneralContractor` Schema:** Provides official address (Accra, Ghana), phone (+233549074200), opening hours, geographic coordinates, service catalog, and logo URLs.
- **`BreadcrumbList` Schema:** Provides clean breadcrumb navigation paths in Google search results.

2. Multi-Platform Favicon & App Icon Matrix

When users open the site in a browser, bookmark it, or add it to their phone's home screen, the browser loads the appropriate asset from `/public`:

- **Browser Tab Favicon:** Loads `favicon-32x32.png`, `favicon-16x16.png`, and `favicon.svg`.
- **iOS Home Screen App:** Loads `apple-touch-icon.png` (180x180) configured via `- **Android & Desktop PWA:** Loads `icon-192x192.png` and `icon-512x512.png` defined in `site.webmanifest` and `manifest.json`.

8. CUSTOMIZATION RECIPES ('HOW TO CHANGE ANYTHING')

Step-by-step developer cheat sheet

Recipe 1: How to Change Phone Number, Email, or WhatsApp

Open `src/config/site.ts` and locate the `company` object around line 10. Update the values:

```
export const company = {  
  // Update your display phone number  
  phone: "0549074200",  
  // Update international dial (no + or leading zero)  
  phoneInternational: "233549074200",  
  // Update email where quotes and contact forms will be sent  
  email: "your-new-email@gmail.com",  
  // Update WhatsApp recipient number  
  whatsapp: "233549074200",  
};
```

Recipe 2: How to Add a New Project to Portfolio

Open `src/config/site.ts`, find the `projects` array, and add a new project object:

```
// Add to the projects array in src/config/site.ts:  
{  
  id: "luxury-residence-cantonments",  
  slug: "luxury-residence-cantonments",  
  title: "Modern 5-Bedroom Executive Villa",  
  category: "Residential",  
  location: "Cantonments, Accra",  
  year: "2025",  
  duration: "10 Months",  
  summary: "Turnkey luxury residential development featuring reinforced concrete framing and  
custom glass finishes.",  
  scope: ["Reinforced Foundation", "Blockwork & Plastering", "Porcelain Tiling", "Swimming Pool"],  
  image: "https://images.unsplash.com/photo-1600585154340-be6161a56a0c?  
auto=format&fit=crop&w=1200&q=80",  
  featured: true,  
},
```

Recipe 3: How to Edit or Add a Construction Service

Open `src/config/site.ts`, find the `services` array, and edit the deliverables, description, or title:

```
// Edit items in the services array in src/config/site.ts:  
{  
  id: "residential-construction",  
  title: "Residential Building Construction",  
  shortTitle: "Residential",  
  icon: "home",  
  summary: "Custom residential villas, duplexes, and family homes built to international  
structural standards.",  
  deliverables: [  
    "Architectural drawing review & structural optimization",  
    "Reinforced concrete slab and column construction",  
    "Precision masonry and premium interior plastering",  
  ],  
},
```

Recipe 4: How to Change Hero Banners or Homepage Taglines

Open `src/config/site.ts` and modify `hero` or `images`:

```
export const hero = {
  eyebrow: "PREMIER BUILDING CONTRACTORS • GHANA",
  title: "Structural Integrity. Architectural Precision.",
  subtitle: "Winnet Construction Ltd delivers residential villas, commercial structures...",
  stats: [
    { value: "100%", label: "Structural Compliance" },
    { value: "15+", label: "Completed Projects" },
  ],
};
```

9. BUILD, DEPLOYMENT & SERVER OPERATIONS

Deploying to production environments

Local Development Commands

```
# 1. Install all dependencies
npm install

# 2. Start local development server (http://localhost:3000)
npm run dev

# 3. Regenerate all PNG favicons & app icons
node scripts/generate-icons.js

# 4. Re-compile this master PDF documentation
node scripts/generate-master-pdf.js

# 5. Build for production
npm run build

# 6. Preview production build locally
npm run preview
```

Deployment Instructions

- **Deploying to Vercel:** Import your GitHub repository into Vercel. Framework preset is automatically detected. Set build command to ``npm run build`` and output directory to default.
- **Deploying to Cloudflare Pages:** Connect repository to Cloudflare Pages, select Vite / TanStack Start preset, set build command ``npm run build``.
- **Deploying to Custom Linux VPS (Ubuntu/Debian):** Run ``npm run build`` followed by process manager like PM2 (``pm2 start npm --name 'winnet' -- run start``).

🔍 Google Search Console Verification

The site includes ``/google4cfcc257bdd58937.html`` and matching ``<meta name="google-site-verification" ...>`` tags in ``src/routes/__root.tsx``. Once deployed to your domain, visit Google Search Console and click 'Verify' to instantly claim your domain and submit your sitemap (``/sitemap.xml``).

